

Longkang Dai

Gender: Male

Nationality: Chinese

Date of Birth: 23 August 1999

Address:

Singapore: Department of the Built Environment, National University of Singapore, 2 Architecture Drive, Singapore, 117566

China: School of Civil Engineering Chongqing University, Chongqing, 400044, China

Telephone: (+86) 13308337433

E-mail: 20221601006@stu.cqu.edu.cn



EDUCATION

- Ph. D candidate, School of Civil Engineering, Chongqing University (Sep. 2022 ~ present)
- Visiting Ph.D. Student, College of Design and Engineering, National University of Singapore (Aug. 2025 - Present)
- Master's Degree student of Civil Engineering, Chongqing University (Sep. 2021 ~ Sep. 2022)
- B.E. in Built Environment and Energy Application Engineering, Chongqing University (Jul. 2021)

PUBLICATIONS

Journal Articles:

1. **L. Dai**, Z. Li, X. Chen, Y. Zhai, C. Fu, Z. Liu, M. Liu*, Usage behavior characteristics of household air-conditioners during the extremely hot summer – A case study of Chongqing, Building and Environment 234 (2023) 110160.
2. M. Yu, Z. Tang, Y. Tao, L. Ma, Z. Liu, **L. Dai**, H. Zhou, M. Liu*, Z. Li, Thermal comfort prediction in multi-occupant spaces based on facial temperature and human attributes identification, Building and Environment 262 (2024) 111772.
3. X. Chen, Z. Li, Z. Liu, **L. Dai**, Q. Fan, J. Li, D. Weng, M. Liu*, A spatial-temporal coefficient method for analyzing the rural dwellings renovation with intermittent heating in China, Energy and Buildings 323 (2024) 114820.
4. X. Chen, Z. Li, **L. Dai**, W. Zeng, Z. Liu, M. Liu*, Characteristics of intermittent heating by air source heat pump in rural areas of northern China based on indoor temperature effective guarantee rate, Journal of Building Engineering (2024) 110196.
5. X. Gu, M. Liu*, Z. Li, H. Liu, X. Chen, **L. Dai**, Dynamic thermal demand analysis of residential buildings based on IoT air conditioner, Building and Environment 243 (2023) 110593.
6. Y. Zhai, Z. Li, X. Qiu, L. Zhu, Y. Lei, **L. Dai**, X. Xie, C. Sekhar, M. Liu*, Impact of transient temperature on the Elderly's physiology and psychology during getting-up process in winter, Building and Environment 253 (2024) 111345.
7. X. Chen, Z. Li, **L. Dai**, M. Liu*, Baseline load characteristics of transitioning to air-to-air heat pump electric heating in rural areas of northern China, Indoor and Built Environment (2024)

1420326X241270825.

8. X. Chen, Z. Li, **L. Dai**, W. Zeng, M. Liu*, Occupant Heating Patterns of Low-Temperature Air-to-Air Heat Pumps in Rural Areas during Different Heating Periods, Buildings 13 (2023) 679.
9. Y. Zhai, L. Ma, M. Liu*, Y. He, C. Zhou, X. Xie, **L. Dai**, H. Liu, Analysis of Regulation Characteristics in Acceptable Thermal Comfort Zone under Cooling Conditions, in: EDP Sciences, 2022: p. 03002.

Academic Conference:

Different using behavior of household air-conditioners in the hotter Summer — an example in Chongqing, Oral talk, The 11th International Conference on Sustainable Development in the Building and Environment(SuDBE 2023), 2023, Helsinki, Finland.

RESEARCH PROJECTS

- A Non-Invasive Perception-Based Method for Predicting Thermal Comfort of People in Airport Terminals Using Time Series Analysis(Grant No: CYB240070) ,2024~2026, the 2024 Chongqing University Graduate Student Research Innovation Project, principal investigator
- Research on Low-Grade Energy Heating Technology for Rural Towns(Grant No: 2018YFD1100704), 2018~2023, the National Key R&D Program of China, principal investigatory student.
- Ensuring health and comfort and realizing low-carbon construction in confined space environments(Grant No: 2024CDJYXTD-003), 2024~2026, the Fundamental Research Funds for the Central Universities, principal investigatory student

PRIZES

- Class A academic scholarship, Chongqing University, 2022 & 2023 & 2024
- China Scholarship Council – PhD Fellowship for Innovative Talents International Cooperation Training Program, ChinaScholarship Council, 2025
- Excellent Undergraduate Graduation Project, Chongqing University, 2021
- Outstanding Student of Chongqing University, Chongqing University, 2025

PATENT AND DATA INTELLECTUAL PROPERTY

- Hanlong Liu, Huan Liu, Wengang Zhang, **Longkang Dai**, Dan Huang, Yuting Pan. Heat insulation device for indoor high and spacious areas, March 18, 2025. China, CN118029559B (Patent Grant)
- Meng Liu, **Longkang Dai**, Zhilong Liu, Shikeng Chen. Typical Residential Air Conditioning Usage Behavior and Energy Consumption Dataset under Extreme High Temperature Conditions, December 9, 2025. China, SZ2025140021378.2

RESEARCH INTERESTS

Embodied intelligence in the built environment; Thermal comfort; Big data mining in buildings; Energy efficiency of RACs; Building Environment High-quality Small Data